

ForceDelta-VLA: Supplementary Material

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I. TRAINING AND ARCHITECTURE DETAILS

The teacher is trained for 50,000 steps with AdamW at a peak learning rate of 2×10^{-5} . It predicts $H=50$ actions at 30 Hz from the latest 100 ms of the 100 Hz wrench stream, and flow-matching inference uses $N_{\text{flow}}=10$ steps.

After freezing the teacher, we train the rank-32 force-agnostic adapter for 10,000 steps at a learning rate of 10^{-4} . The visual-language prefix is pooled into $M=16$ positional bins and projected to $Q=2$ task-context tokens. The correction policy comprises a causal force encoder and one pre-norm set-attention block with separate zero-initialized force and delay-correction output projections. It predicts $K=5$ corrections and is trained for 30,000 steps at a learning rate of 3×10^{-4} under the sampled asynchronous schedule. The delay-correction pass masks the force-token position in the attention-validity mask.

A. Action Representation and Context Encoding

Pose actions use $\mathcal{P}(A) = [p; r]$, where $p \in \mathbb{R}^3$ is translation and $r \in \mathbb{R}^6$ is the continuous rotation representation. Pose actions and wrench measurements use the corresponding robot-base frame. Differences are scaled by the training-set action standard deviations. Rotational corrections are added in the teacher’s 6D action coordinates and mapped to valid rotation matrices by Gram–Schmidt orthogonalization. Bimanual inputs and outputs concatenate the per-arm quantities; correction construction is component-wise and bounds are applied independently to each arm.

The visual-language prefix E_k excludes state, force, and action tokens. Its mask m_k selects available image-view and non-padding language tokens. Masked mean pooling partitions the padded prefix into $M = 16$ contiguous, approximately equal-width positional bins:

$$(\bar{E}_k, \bar{m}_k) = \text{Pool}_M(E_k, m_k), \quad Z_k^{\text{ctx}} = P_{\text{ctx}}(\bar{E}_k, \bar{m}_k). \quad (1)$$

The learned projector uses $Q = 2$ queries to aggregate valid pooled tokens. Condition-specific projections map task context, force history, state, reference-action segment, and timing features to the shared token width. Learned type embeddings identify the conditions; no positional encoding is used in the conditioning set. Permuting condition tokens together with their type embeddings leaves the fixed correction-query output invariant. The force and delay passes share the attention block and use separate output projections; the delay pass masks the force token.

Cache age and interpolation phase are represented as

$$\xi_{t,k} = \left[\min\left(\frac{t - t_k}{c}, 1\right); \alpha_{t,k} \right], \quad c = 0.6 \text{ s}, \quad (2)$$

where $\alpha_{t,k}$ is the position of t between adjacent reference-action samples.

The action representation and rotational correction composition are defined in Sec. III-A of the main paper. Let $\tilde{S}^p$ be the nine per-arm pose coordinates of the normalized robot state, obtained after converting orientation to the continuous 6D representation, and let σ_S^p and σ_A^p be the corresponding training-set state and action standard deviations. The implementation computes

$$\Gamma(S_t, S_k) = \frac{\sigma_S^p + \varepsilon}{\sigma_A^p + \varepsilon} \odot (\tilde{S}_t^p - \tilde{S}_k^p), \quad \varepsilon = 10^{-6}. \quad (3)$$

This scale-corrected componentwise difference is included in the learned delay-correction target, placing the current force-agnostic prediction and reference action in the same normalized delta-action coordinate system. The composed normalized action is de-normalized, applied relative to the reference-action-query state S_k , and projected to a valid rotation by Gram–Schmidt orthogonalization.

The correction policy observes the complete K -step reference-action segment aligned with its output timestamps. The segment is flattened and projected to one typed reference-action token, so each predicted correction is conditioned on the corresponding future reference-action waypoint without increasing the attention-sequence length.

At deployment, force and delay corrections are clipped separately and element-wise before composition. The implementation accepts either nine-dimensional nonnegative bounds $b_F, b_S \in \mathbb{R}_{\geq 0}^9$ or scalar values broadcast over the pose coordinates; bimanual clipping is applied independently per arm. **The final robot configuration must specify the values used for b_F and b_S .**

Correction-policy training examples are anchored at the 30 Hz state/action timestamps while retaining every 100 Hz wrench sample in $\mathcal{F}_t$. The sampled schedule independently draws inter-query periods $T_i \sim \mathcal{U}(1/15, 1/5)$ s and reference-action-query latencies $L_i \sim \mathcal{U}(0.05, 0.30)$ s. Query rows are first selected greedily on the recorded grid, after which query timestamps receive $J_i \sim \mathcal{U}(-0.01, 0.01)$ s jitter to exercise off-grid reference-action interpolation. Correction queries may start more frequently at deployment; the realized correction-update rate is measured directly rather than inferred from the action-sample rate within a chunk.

B. Force-Agnostic Mode Optimization

The learned force-agnostic token and mode-specific adapter are active only for force-agnostic queries. With the force-conditioned teacher parameters θ fixed, the reference-action

parameters ϕ are optimized using the teacher's original flow-matching objective,

$$\mathcal{L}_{\text{ref}} = \mathbb{E}_{\mathcal{D}, \eta, \epsilon} [\ell_{\text{FM}}(T_{\theta, \phi}^{\text{ref}}, A_{t:t+H-1}^*; \eta, \epsilon)], \quad (4)$$

where η is flow time and ϵ is the initial flow noise. Force-conditioned and force-agnostic predictions used to construct a correction target share the same ϵ .

C. Correction-Loss Weighting

For a K -step correction chunk, the normalized temporal weights are

$$w_j = \frac{\beta^{j-1}}{\sum_{m=1}^K \beta^{m-1}}, \quad 0 < \beta \leq 1. \quad (5)$$

Thus, $\beta < 1$ emphasizes near-term corrections, $\beta = 1$ weights all steps equally, and $\sum_j w_j = 1$ keeps the loss scale stable as K changes. The loss is a mean-squared error in normalized pose-action coordinates.

II. ASYNCHRONOUS REFERENCE-ACTION AND CORRECTION EXECUTION

A. Reference-Action Updates

A reference-action query k , started at t_k from state S_k , publishes the immutable reference-action update $(A_{T,k,1:H}^{\text{ref}}, \bar{E}_k, \bar{m}_k, S_k, t_k, \bar{t}_k)$ when its result becomes available at $\bar{t}_k$. Its samples carry intended execution times

$$t_{k,n}^{\text{ref}} = t_k + (n-1)/f_A, \quad n = 1, \dots, H. \quad (6)$$

At time t , the active reference action is the freshest completed, non-superseded query,

$$k^*(t) = \arg \max_{k: \bar{t}_k \leq t} t_k. \quad (7)$$

An older query that completes later therefore cannot replace a newer active reference-action update. Samples whose intended times precede $\bar{t}_{k^*}$ are skipped for direct execution; the cached chunk itself is not truncated. For $t \geq \bar{t}_{k^*}$, the reference pose $A_{k^*}^{\text{ref}}(t)$ is obtained by timestamp-based interpolation over the original timestamped chunk.

B. Correction Updates

A correction query q , started at t_q^F , reads $k_q = k^*(t_q^F)$ together with $(\mathcal{F}_{t_q^F}, S_{t_q^F})$. It completes at $\bar{t}_q^F$ with force-conditioned and delay correction chunks. Their samples are timestamped as

$$t_{q,j}^F = t_q^F + (j-1)/f_A, \quad j = 1, \dots, K. \quad (8)$$

Here f_A defines the spacing between correction samples within a chunk; correction query start times t_q^F are scheduled independently. Step j is held over $[t_{q,j}^F, t_{q,j}^F + 1/f_A)$, and correction samples are not interpolated. At control time t , the executor selects the most recently started correction query whose result has completed and remains valid, together with its active step:

$$q^*(t) = \arg \max_{q: \bar{t}_q^F \leq t < \bar{t}_q^F + K/f_A} t_q^F, \quad (9)$$

$$j(t; q^*) = 1 + \lfloor (t - t_{q^*}^F) f_A \rfloor.$$

Any prefix that expires during inference is therefore skipped, while the remaining valid steps execute sequentially until a newer correction chunk completes. Let $j^* = j(t; q^*)$. Within the selected interval, Eq. (1) of the main paper is evaluated using the selected correction query time and active step:

$$\begin{aligned} \mathcal{P}(A^{\text{cmd}}(t)) &\equiv \mathcal{P}(A_{t_{q^*}^F, j^*}^{\text{cmd}}) \\ &= \mathcal{P}\left(A_{k_{q^*}}^{\text{ref}}(t_{q^*}^F, j^*)\right) + \Delta \hat{A}_{q^*, j^*}^{\text{force}} + \Delta \hat{A}_{q^*, j^*}^{\text{delay}}. \end{aligned} \quad (10)$$

The two corrections are clipped independently and element-wise in normalized delta-action coordinates before composition. The composed command remains referenced to $S_{k_{q^*}}$ and is converted to an absolute robot command as described above. A correction chunk that completes after $t_q^F + K/f_A$ is rejected. If no valid correction chunk remains, the executor invokes the platform's safe hold or abort policy. **The final implementation record will specify gripper sampling and behavior beyond the last valid reference-action timestamp.**

III. ADDITIONAL EVALUATION PROTOCOLS

A. Task Success Criteria

The task-specific success criteria are summarized in Table I.

B. Metric Aggregation

As in the main paper, peak force and completion time are reported as mean $\pm$ standard deviation over successful trials pooled across tasks within each platform. Consequently, a task's contribution to the pooled metrics depends on the method's number of successes on that task. Success rate is averaged across the nine tasks, with 20 trials per method and task. **The precise per-trial force reduction across arms and completion-time boundaries must be verified against the final evaluation code.**

C. Task-Wise Trial Counts

The final experiment record will report the successful-trial count for each method and task. Each count is out of twenty evaluation trials and aggregates to the main paper's success-rate table.

TABLE I
TASK SUCCESS CRITERIA. RED ENTRIES ARE PROVISIONAL THRESHOLDS.

Platform	Task	Success criterion
Single-arm	Object Flipping	Object rotated by at least 150° and left stable
Single-arm	USB Insertion	Connector seated to within 2 mm of its terminal depth
Single-arm	Cabinet Opening	Door opened beyond 60°
Single-arm	Button Pressing	Button fully actuated
Single-arm	Whiteboard Wiping	At least 90% of the marked region removed
Bimanual	Plug Removal	Plug fully separated without dropping either component
Bimanual	Plug Insertion	Plug fully seated while the second arm stabilizes the socket
Bimanual	Drawer Opening	Drawer translated by at least 15 cm
Bimanual	Whiteboard Wiping	At least 90% of the marked region removed while the left arm stabilizes the board and the right arm wipes